

Audio and Video Authentication for Robots

1. Domain

- Artificial Intelligence
 - Robotics
 - Computer Vision
 - Speech Processing
 - Cybersecurity
 - Human-Robot Interaction
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2. Problem Statement

As robots become increasingly common in homes, hospitals, industries, warehouses, and research laboratories, ensuring that only authorized users can access or control them has become a critical security challenge. Most existing robotic systems rely on passwords, RFID cards, or mobile applications, which are susceptible to theft, misuse, or spoofing.

There is a need for an intelligent authentication system capable of accurately verifying a user's identity before granting access to the robot. A multimodal biometric authentication system that combines facial recognition and voice verification provides a more secure, reliable, and user-friendly solution while minimizing unauthorized access.

3. Project Description

The **Audio and Video Authentication for Robots** project aims to develop an AI-powered multimodal authentication system that verifies users through facial recognition and voice authentication before allowing interaction with a robot.

The system captures facial images using a camera and voice samples through a microphone. Computer Vision algorithms identify the user's face, while Speech Processing models verify the speaker's identity. Both authentication results are combined using a decision engine to determine whether the user is authorized.

Only verified users can control or interact with the robot. If authentication fails, the robot denies access, records the event, and optionally alerts the administrator.

The project focuses on improving robotic security, authentication accuracy, and user convenience while reducing the risks associated with single-factor authentication.

4. Project Objectives

- Develop a secure multimodal authentication system.
- Authenticate users using face recognition and voice verification.
- Improve authentication accuracy through biometric fusion.
- Prevent unauthorized access to robotic systems.
- Reduce spoofing attacks compared to single-modal authentication.
- Enable secure human-robot interaction.

- Create a scalable authentication framework for various robotic platforms.

5. Project Scope

The project focuses on developing a secure multimodal authentication system for robots using facial and voice recognition. It includes user enrollment, real-time identity verification, multimodal decision-making, access control, authentication logging, and intrusion detection. The system is designed to enhance robotic security while providing a scalable framework for future integration with advanced AI techniques and robotic platforms. It also establishes a robust foundation for secure and intelligent access management in next-generation robotic systems.

6. Motivation

Robots are increasingly deployed in security-sensitive environments where unauthorized access can result in:

- Data leakage
- Privacy violations
- Equipment misuse
- Operational disruption
- Safety hazards

Using two independent biometric traits significantly enhances security while maintaining ease of use.

7. Existing System

Most robots currently use:

- Passwords
- PINs
- RFID cards
- Mobile applications
- Single-face authentication
- Fingerprint scanners

Limitations

- Passwords can be shared.
- RFID cards can be lost.
- Fingerprints require physical contact.
- Face-only systems are vulnerable to spoofing.
- Voice-only systems are affected by environmental noise.

8. Proposed System

The proposed system combines:

- Face Recognition
- Voice Authentication
- Multimodal Decision Engine

The robot grants access only when authentication criteria are satisfied, providing higher security and improved reliability.

9. Key Features

- Face Detection
- Face Recognition
- Voice Authentication
- Multimodal Authentication
- Real-Time Verification
- User Enrollment
- Access Control
- Authentication Logging
- Intruder Detection
- Alert Generation
- User Database Management

10. Functional Modules

Module 1: User Enrollment

- Register authorized users
- Capture multiple face images
- Record voice samples
- Generate biometric embeddings
- Store encrypted templates

Module 2: Face Recognition

Functions:

- Face Detection
- Face Alignment
- Face Embedding
- Identity Matching

Module 3: Voice Authentication

Functions:

- Voice Recording
- Noise Reduction

- Feature Extraction
- Speaker Verification

Module 4: Authentication Engine

Combines:

- Face confidence score
- Voice confidence score

Module 5: Robot Access Control

Authorized User → Grant Access → Accept Commands

Else → Reject Access → Alert Administrator

Module 6: Security Logging

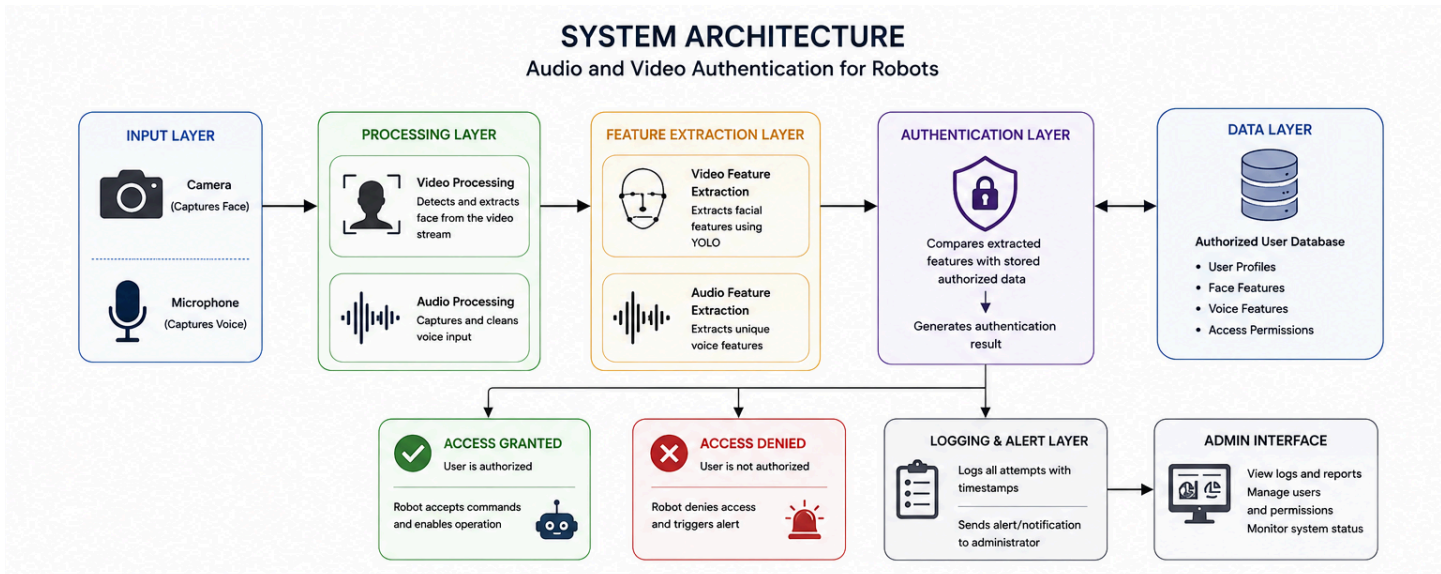
Stores:

- User ID
- Date
- Time
- Authentication Result
- Confidence Score
- Captured Image
- Audio Sample
- Device Information

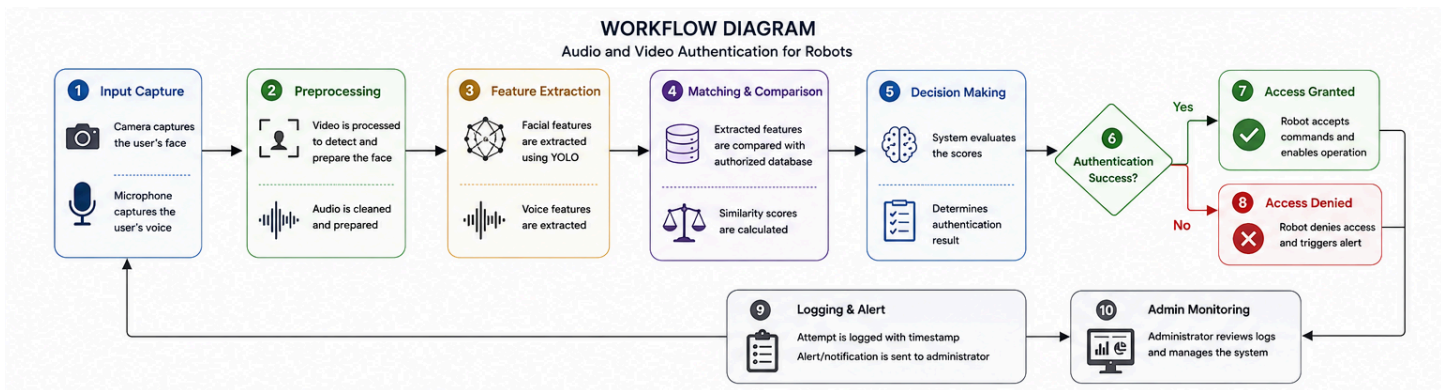
Module 7: Database Management

- Store enrolled users
 - Update biometric templates
 - Manage user permissions
 - Maintain authentication logs
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11. System Architecture



12. Workflow



13. Technology Stack

Programming Language

- Python

Computer Vision

- OpenCV

Face Detection

- YOLO

Face Recognition

- FaceNet / ArcFace / InsightFace

Speaker Recognition

- MFCC (Feature Extraction)
- Speaker Verification Model

Audio Processing

- Librosa
- SpeechRecognition

Deep Learning

- PyTorch

Database

- SQLite
 - PostgreSQL
 - MongoDB
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14. Security Features

- Multimodal Authentication
 - Confidence Score Fusion
 - Encrypted Biometric Templates
 - Secure Database Storage
 - Authentication Logging
 - Role-Based Access Control
 - Session Timeout
 - Access History
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15. Applications

- Home assistant robots
- Healthcare robots
- Industrial automation
- Warehouse robots
- Research laboratories
- Educational robots
- Military robots

- Defense surveillance
 - Smart factories
 - Hospitality robots
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16. Advantages

- High authentication accuracy
 - Improved security
 - Reduced spoofing risk
 - Contactless authentication
 - User-friendly interface
 - Fast verification
 - Easy integration with robots
 - Scalable architecture
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17. Limitations

- Performance may degrade in poor lighting.
 - Background noise can affect voice recognition.
 - Requires quality camera and microphone.
 - Similar-looking individuals may reduce face recognition accuracy.
 - Changes due to masks, helmets, or illness can impact authentication.
 - Large biometric databases increase matching time.
 - Requires periodic model updates for optimal performance.
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18. Challenges

Technical Challenges

- Face occlusion
- Voice variation due to illness
- Environmental noise
- Camera positioning
- Lighting changes
- Real-time processing constraints
- Synchronizing audio and video streams
- Edge device computational limitations

Security Challenges

- Replay attacks

- Printed photo attacks
 - Deepfake face attacks
 - Voice cloning
 - Database security
 - Network attacks
 - Device tampering
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19. Performance Metrics

- Face Recognition Accuracy
 - Speaker Verification Accuracy
 - Authentication Accuracy
 - False Acceptance Rate (FAR)
 - False Rejection Rate (FRR)
 - Precision
 - Recall
 - F1 Score
 - Authentication Time
 - System Latency
 - Throughput
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20. Future Scope

The project can evolve into a comprehensive intelligent robotic security platform through the following enhancements:

Advanced Biometrics

- Iris recognition
- Fingerprint authentication
- Palm vein recognition
- Gait recognition

Enhanced AI

- Context-aware decision making using Vision-Language Models (VLMs)
- Behavioral analysis to detect suspicious activities
- Continuous authentication while interacting with the robot
- Personalized user profiles and adaptive authentication

Robotics Integration

- Integration with ROS2-based autonomous robots
- Multi-robot authentication management

- Secure remote robot operation
- Human-following robots with owner verification

Security Improvements

- Anti-spoofing using liveness detection (blink detection, challenge-response)
- Deepfake and voice clone detection
- End-to-end encrypted biometric storage
- Blockchain-based audit logs (for high-security environments)

IoT and Smart Environment Integration

- Smart doors and access control systems
- Smart home automation
- Industrial IoT ecosystems
- Cloud-based monitoring dashboards
- Mobile application for administrators

Scalability

- Multi-user access management
- Organization-wide biometric databases
- Distributed authentication across multiple robots
- Cloud synchronization for fleet management

Explainable AI

- Display authentication confidence scores
- Visualize reasons for access denial
- Generate security reports and analytics

21. Expected Outcome

The project will deliver an intelligent multimodal authentication system that enables robots to securely verify users through facial and voice recognition before accepting commands. By leveraging AI-based biometric authentication, the system will enhance security, minimize unauthorized access, improve reliability, and provide a robust foundation for future intelligent robotic security systems. As the project evolves, it can incorporate advanced AI reasoning, anti-spoofing mechanisms, and smart environment integration, making it suitable for deployment in real-world homes, hospitals, industries, research facilities, and autonomous robotic ecosystems.